Nome: Luiz Felipe Ciantela Machado

Turma: CTII-348

Prontuário: CB1990209

Cálculo Geral de Determinantes

(Fotos nas páginas seguintes)

Exercício 1:

D S T Q Q S S

Sarrus Básico - Determinantes - Cálculo geral.

01-)

$$A = \left|\begin{array}{ccc|cc} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & -1 & 1 & 0 & -1 \end{array}\right|$$

$0 - 1 + 0 = \boxed{-1}$

$1 + 0 + 0 = \boxed{1}$

$1 - (-1) = \boxed{2}$

$\boxed{\det A = 2}$

$$B = \begin{vmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & 0 & 3 \\ 0 & 1 & 1 & 4 \end{vmatrix}$$

escolhido

1. cof (a_{22}) → Por

$$1. \left|\begin{array}{ccc|cc} 1 & 0 & 3 & 1 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 \end{array}\right|$$

$0 + 3 + 0 = \boxed{3}$

$0 + 0 + 0 = \boxed{0}$

$0 - 3 = \boxed{-3}$ → $1 \cdot -3 = \boxed{-3}$

1. cof (a_{42}) → Por

$$1. \left|\begin{array}{ccc|cc} 1 & 0 & 3 & 1 & 0 \\ 0 & -1 & 4 & 0 & -1 \\ 0 & 0 & 3 & 0 & 0 \end{array}\right|$$

$0 + 0 + 0 = \boxed{0}$

$-3 + 0 + 0 = \boxed{-3}$

$-3 - 0 = \boxed{-3}$ → $-3 \cdot 1 = \boxed{-3}$

$\det B = -3 + (-3) = \boxed{-6}$

$\boxed{\det B = -6}$

Exercício 2:

02 -> Calcule X.

$$\begin{vmatrix} x^2 & 0 & x & -1/10 \\ 7,5 & 0 & 5 & 2 \\ 10 & 0 & 4 & 2 \\ 1 & 1 & 1 & 1 \end{vmatrix} = 0$$

resolvido

1. Col (a, 4, 2)

$$-5 + 8x^2 + 15x = \boxed{8x^2 + 15x - 5}$$

$$1. \begin{vmatrix} x^2 & x & -1/10 \\ 7,5 & 5 & 2 \\ 10 & 4 & 2 \end{vmatrix} \begin{matrix} x^2 & x \\ 7,5 & 5 \\ 10 & 4 \end{matrix}$$

$10x^2 + 20x - 3 - (8x^2 + 15x - 5)$

$10x^2 + 20x - 3 - 8x^2 - 15x + 5 \Rightarrow$

$\Rightarrow \boxed{2x^2 + 5x + 2}$

$$10x^2 + 20x - 3 = \boxed{10x^2 + 20x - 3}$$

$$\underset{a}{2x^2} + \underset{b}{5x} + \underset{c}{2} = 0$$

$\Delta = 5^2 - 4 \cdot 2 \cdot 2$

$\Delta = 25 - 16$

$\Delta = \boxed{9} \rightarrow \sqrt{9} = \boxed{3}$

$$\frac{-5 \pm 3}{2 \cdot 2} \rightarrow \frac{-\cancel{2}^{1}}{\cancel{4}^{2}} = \boxed{-\frac{1}{2}} \rightarrow x'$$

$$\Rightarrow \frac{-8}{4} = \boxed{-2} \rightarrow x''$$

$$\boxed{x = -2 \text{ ou } x = -1/2}$$

Exercício 3:

Q3 -)

$$\begin{vmatrix} x & 0 & 0 & 3 \\ -1 & x & 0 & 0 \\ 0 & -1 & x & 1 \\ 0 & 0 & -1 & -2 \end{vmatrix}$$

← escolhido

$x \cdot cof(a_{11}) \rightarrow$ par

$$x \cdot \begin{vmatrix} x & 0 & 0 \\ -1 & x & 1 \\ 0 & -1 & -2 \end{vmatrix} \begin{matrix} x & 0 \\ -1 & x \\ 0 & -1 \end{matrix}$$

$0 - x - 0 = \boxed{-x}$

$-2x^2 + 0 + 0 = \boxed{-2x^2}$

$-2x^2 - (-x) = \boxed{-2x^2 + x}$

$x \cdot (-2x^2 + x)$

$\boxed{-2x^3 + x^2}$

$-1 \cdot cof(a_{21}) \rightarrow$ impar

$$-1 \cdot \begin{vmatrix} 0 & 0 & 3 \\ -1 & x & 1 \\ 0 & -1 & -2 \end{vmatrix} \begin{matrix} 0 & 0 \\ -1 & x \\ 0 & -1 \end{matrix}$$

$0 + 0 + 0 = \boxed{0}$

$0 + 0 + 3 = \boxed{3}$

$3 - 0 = \boxed{3} \rightarrow \boxed{-3}$

$\rightarrow -1 \cdot (-3) = \boxed{3}$

$\boxed{det = -2x^3 + x^2 + 3} \rightarrow \boxed{det = -2x^3 + x^2 + 3}$

letra A.

Exercício 4:

Q4-)

$$A = \begin{vmatrix} \boxed{x} & 1 & 0 & 0 & 0 \\ 0 & x & 1 & 0 & 0 \\ 0 & 0 & x & 1 & 0 \\ 0 & 0 & 0 & x & k \\ 0 & 0 & 0 & 1 & x \end{vmatrix}$$

↳ escolhido

$f(x) = \det A$ e $f(-2) = 8$

$x \cdot \text{cof}(1,1) \rightarrow$ Laplace

$$x \cdot \begin{vmatrix} \boxed{x} & 1 & 0 & 0 \\ 0 & x & 1 & 0 \\ 0 & 0 & x & k \\ 0 & 0 & 1 & x \end{vmatrix}$$

$x \cdot \text{cof}(a_{11}) \rightarrow$ Laplace

$$x \cdot \begin{vmatrix} x & 1 & 0 \\ 0 & x & k \\ 0 & 1 & x \end{vmatrix} \begin{matrix} x & 1 \\ 0 & x \\ 0 & 1 \end{matrix}$$

$0 + k \cdot x + 0 = \boxed{kx}$

$\boxed{x^3 - kx}$

$x^3 + 0 + 0 = \boxed{x^3}$

$x \cdot (x^3 - kx)$

$\boxed{x^4 - kx^2}$

$x \cdot (x^4 - x^2 \cdot k) \Rightarrow \boxed{x^5 - x^3 k} \Rightarrow$ letra

$f(x) = \det A \Rightarrow f(x) = x^5 - x^3 k$ ∴

$\therefore f(-2) = -2^5 - (-2)^3 k = 8$

$f(-2) = -32 + 8k = 8$

$-32 + 8k = 8$

$-8k = -32 - 8 \quad (\cdot -1)$

$8k = 32 + 8$

$8k = 40$

$k = \frac{40}{8} \Rightarrow \boxed{k = 5}$ $\boxed{k = 5}$ letra D